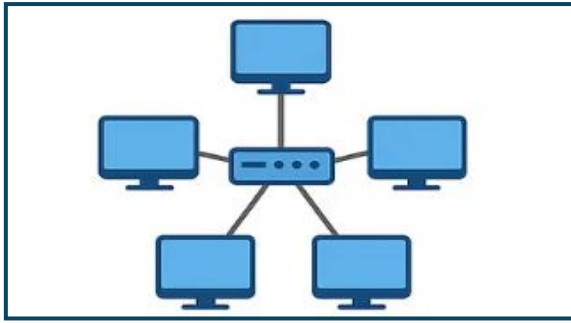


1 Star Topology



In a **Star Topology**, all computers, laptops, or other end devices are connected separately to one **central device**, such as a **switch or hub**.

Data flow:

Computer A → Central Switch/Hub → Computer B

Real-World App/Service Example: WhatsApp :

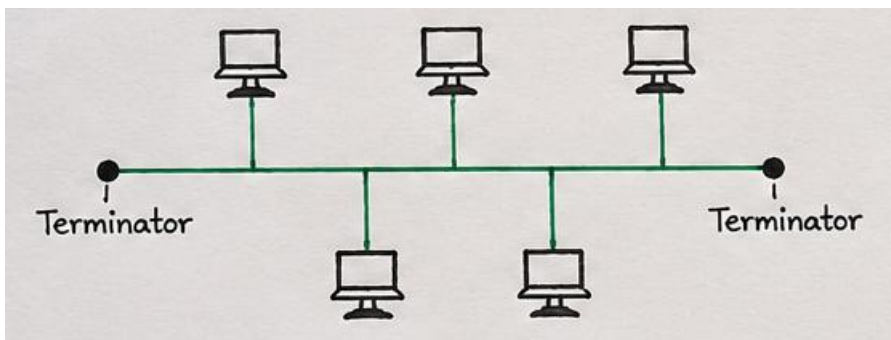
WhatsApp can be used as a simplified example because users communicate through a centrally managed service infrastructure. When one user sends a message, the service handles delivery to the intended recipient rather than users forming a direct physical cable connection with each other.

For example:

Sender → WhatsApp Service Infrastructure → Receiver

This is conceptually similar to star topology because communication depends on central service infrastructure, just as devices in a star network depend on a central switch or hub.

2 Bus Topology



In a **Bus Topology**, all computers and devices are connected to one main communication cable called the **backbone cable** or **bus**.

Basic structure:

Terminator — PC — PC — PC — PC — Terminator

A **terminator** is placed at both ends of a traditional bus network to reduce signal reflection.

Real-World App/Service Example: IPL Streaming :

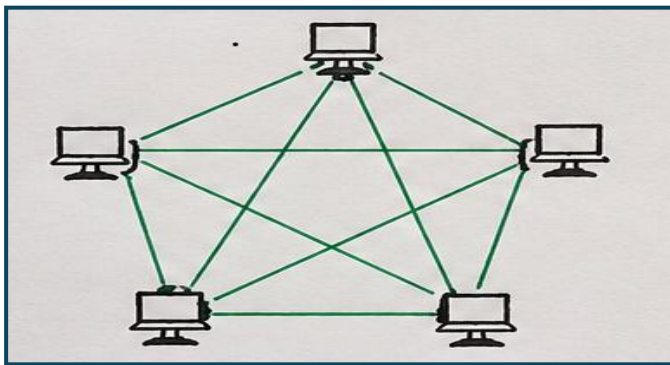
IPL streaming can be used only as a simplified analogy. A live cricket stream is distributed to a very large number of viewers, so the idea of shared or broadly distributed content can help students visualize bus-style communication.

For example:

Live Match Source → Distribution Network → Many Viewers

This resembles the broad idea of information being distributed to many connected receivers. However, actual modern IPL streaming platforms use complex internet infrastructure, including servers and content delivery networks, rather than a literal bus topology.

3 Mesh Topology



In a **Mesh Topology**, devices or network nodes are connected through multiple paths. In a **full mesh topology**, every node has a direct connection to every other node.

For example, if there are four devices A, B, C, and D:

- A connects to B, C, and D.
- B connects to A, C, and D.
- C connects to A, B, and D.
- D connects to A, B, and C.

Because multiple paths are available, data may still reach its destination even if one connection fails.

Real-World App/Service Example: Flipkart

Flipkart can be used as a simplified conceptual example of a large distributed online service. Large e-

commerce systems may depend on many interconnected services, servers, network routes, and infrastructure components.

If one component or route fails, another available component or route may help keep the service running. This resembles the main idea of mesh topology: **multiple paths and redundancy**.

For example:

User → Server/Service A → Database/Service B

If one path is unavailable:

User → Server/Service C → Alternative Backend Path

This improves reliability and availability, especially during heavy traffic such as major online sales.